

ELEN 3201 Circuit Analysis Experiment 3

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1 Part 1

1.1 (a)

$$\begin{aligned}i_c &= C \frac{dV_c}{dt} \\ \frac{V_1 - 0}{R_1} &= C_1 \frac{dV_c}{dt} = C_1 \frac{d}{dt} (0 - V_a(t)) \\ \frac{V_1}{R_1} &= -C_1 \frac{d}{dt} V_a(t) \\ V_a(t) &= -\frac{1}{R_1 C_1} \int_0^t V_1(\tau) d\tau \\ &= -\frac{V_1}{R_1 C_1} t + V_a(0) \\ &= -\frac{V_1}{R_1 C_1} t \\ &= -\frac{1}{5} V_1 t\end{aligned} \tag{1}$$

1.2 (b)

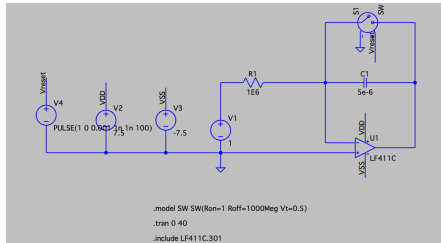
Switch closed: the capacitor is shorted and the voltage across it is zero. Therefore,

$$V_a(t) = 0$$

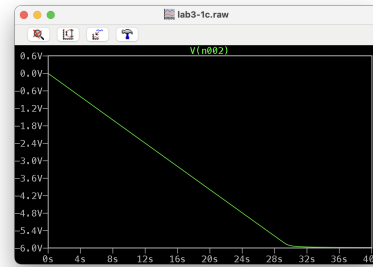
1.3 (c)

Verify with LTspice for $R_1=1\text{M } \Omega$ and $C_1=5\mu\text{F}$:

$$V_a(t) = -\frac{V_1}{R_1 C_1} t = -\frac{1}{5} V_1 t$$



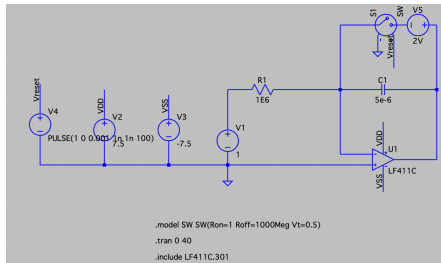
(a) Part 1(c) LTspice schematic



(b) LTspice simulation result

Figure 1: Part 1(c) schematic and simulation

1.4 (d)



(a) Part 1(d) Diagram



(b) Part 1(d) Simulation

Figure 2: Part 1(d) simulation and diagram



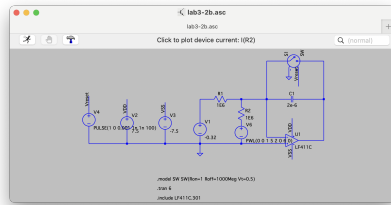
Figure 3: Part1 d Measurement

2 Part 2

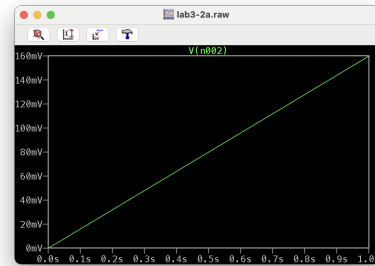
$$\begin{aligned} \text{Input (a): } 100 \text{ ft/s}^2 &\rightarrow 1V \\ \text{Output (v): } 200 \text{ ft/s} &\rightarrow 1V \end{aligned} \quad (2)$$

2.1 (a)

Given the scales above, when $t = 1s$, the velocity $v = at = -32 \text{ ft/s}^2 * 1s = -32 \text{ ft/s}$. This velocity is represented by $-0.16V$, as our SPICE simulation result shown in Figure 4b.



(a) Part 2(a) diagram



(b) Result part 2(a)

Figure 4: Part 2(a) result and diagram

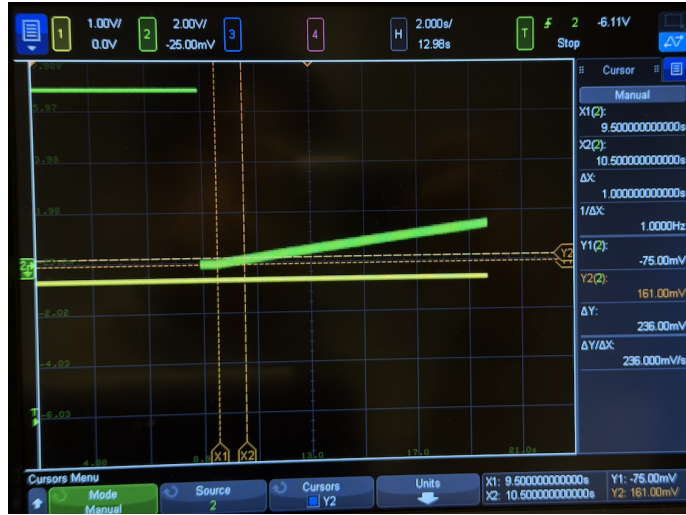


Figure 5: Part 2 a measurement

2.2 (b)

t (s)	a (ft/s ²)	Δv (ft/s)
0-1	$500t - 32$	218
1-2	$-500(t - 2) - 32$	218
2-6	-32	-128

So, the entire kinematics from 0-6 seconds are as follow,

$$v(t = 6 \text{ s}) = 218 + 218 - 128 = 308 \text{ ft/s.}$$

$$308 \text{ ft/s} \longleftrightarrow -1.54 \text{ V} \quad \text{at } t = 6 \text{ s.}$$

SPICE result (Figure 7) matches our calculation. For instance, at $t = 6 \text{ s}$, voltage is -1.54 V

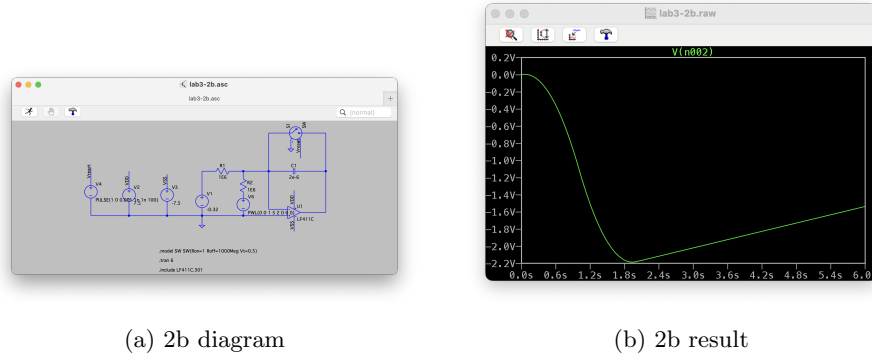


Figure 6: Part 2 b Diagram and Result



Figure 7: Part 2 b Measurement

During the lab, we cannot realize the perfect time-varying acceleration start at zero, rise linearly to 500f/s^2 by $t = 1\text{sec}$, then fall linearly back to zero at $t = 2\text{sec}$. The measurement matches the simulation in Figure 6b qualitatively where the minimum is at approximately 2 s.

3 Part 3

3.1 (a)

According to the Equation 2, the velocity 100ft/s is represented by -0.5V . After 10 seconds, the distance became 1000ft , represented by 1V according to the scale of $1\text{V} = 1000\text{ft}$.

We know that the integrator formula is $V_{\text{out}} = -V_{\text{in}} \cdot \frac{1}{RC}t$ as shown in the part 1 a). To calculate the distance based on this voltage correspondence, we need to pick $RC=5$ such that $V_{\text{out}} = -V_{\text{in}} \cdot \frac{1}{RC} \cdot 1\text{sec} = -1\text{V}$. Therefore, we chose $1\mu\text{F}$ capacitor and $5\text{M}\Omega$ resistor. What is the final output voltage and what distance does that represent? The final output voltage is -1V, this represents 1000ft.

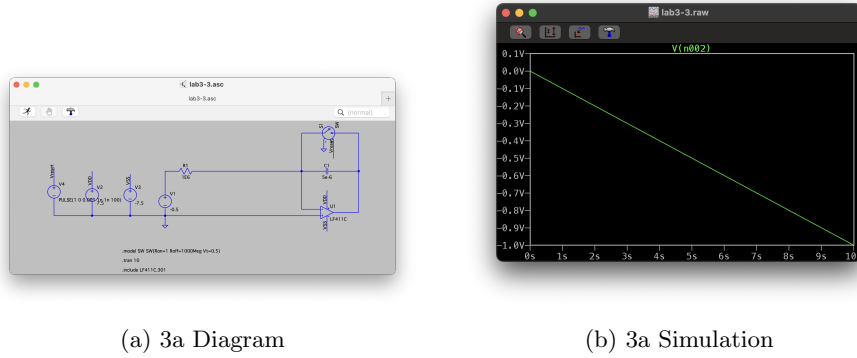


Figure 8: Part 3(a) diagram and result



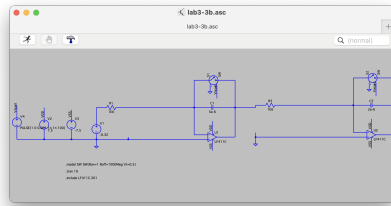
Figure 9: Part3a Measurement

3.2 (b)

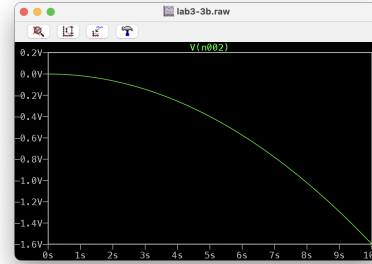
What is the velocity and distance traveled for a body passively falling for 10 seconds?

velocity: $v = gt = 320 \text{ ft/s} \longleftrightarrow 1.6 \text{ V}$

displacement: $x = \frac{1}{2}gt^2 = 1600 \text{ ft} \longleftrightarrow -1.6 \text{ V}$



(a) 3b diagram



(b) 3b result

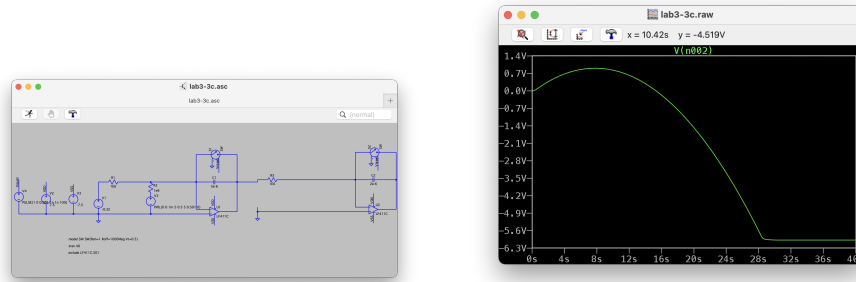
Figure 10: Part 3(b) diagram and result



Figure 11: 3b Measurement, 8s, 800ft, calculated 7.8s, 980ft

3.3 (c)

Include provision for a non-zero initial velocity



(a) 3c diagram

(b) 3c result

Figure 12: Part 3(c) diagram and result

3.4 (d)

What is the projectile's maximum height? At what time does the simulation cease to represent the physical situation? Why?

Kinematic:

$$t_{\text{peak}} = \frac{250}{32} = 7.8 \text{ s}$$

$$h_{\text{max}} = \frac{v_0^2}{2g} = \frac{255^2}{64} = 980 \text{ ft} \leftrightarrow 0.98\text{V}$$

$$t_{\text{hit ground}} = \frac{2v_{\text{initial}}}{g} = \frac{500}{32} = 15.6 \text{ s}$$

Setting a 5 V pulse for 0.5 seconds , i.e. PWL(0 0 1m 5 0.5 5 0.501 0), we can obtain a initial velocity of 250 ft/s



Figure 13: Part 3 d Measurement. Matches calculated 980ft at 7.8s

The simulation ceases to represent the physical situation when the rocket returns the ground ($y=0$) because our model did not consider this boundary condition.

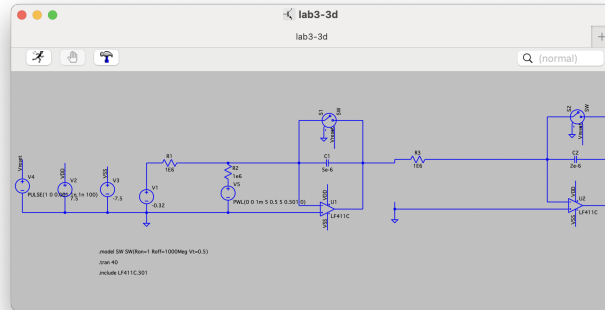


Figure 14: Part 3 d Diagram

4 Part 4

What is the maximum height, velocity, and duration of the flight?

$$F = ma$$

$$a_{up} = \frac{T(t)}{m}$$

$$V_{\text{thrust}} = \frac{a}{100} = \frac{T(t)}{0.2} \cdot \frac{3.28}{100} = T(t) \cdot 0.164 \text{ V}$$

Applying this formula to obtain the voltage measurement of the thrust, we can get data points for the PWL. The result is attached in the appendix.

Maximum height: $\approx 225.78 \text{ ft}$

Maximum velocity: $79.8 \text{ mV} \times 200 \text{ ft/s per V} \approx 15.96 \text{ ft/s}$

Duration of flight: $\approx 10 \text{ s}$

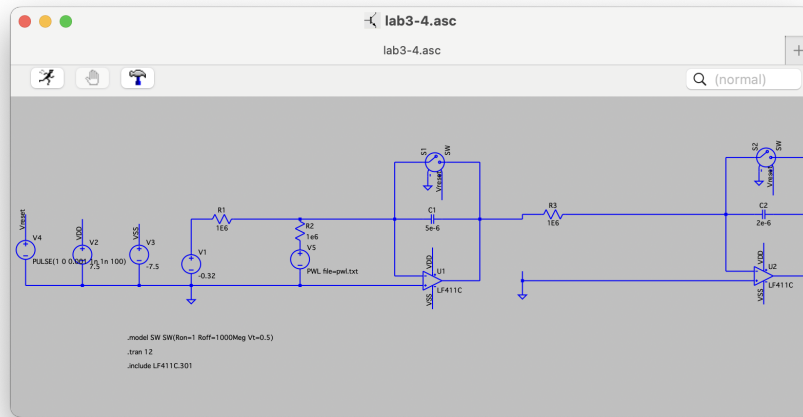


Figure 15: Part 4 diagram



Figure 17: Part 4 Maximum velocity: $79.8 \text{ mV} \times 200 \text{ ft/s per V} \approx 15.96 \text{ ft/s}$

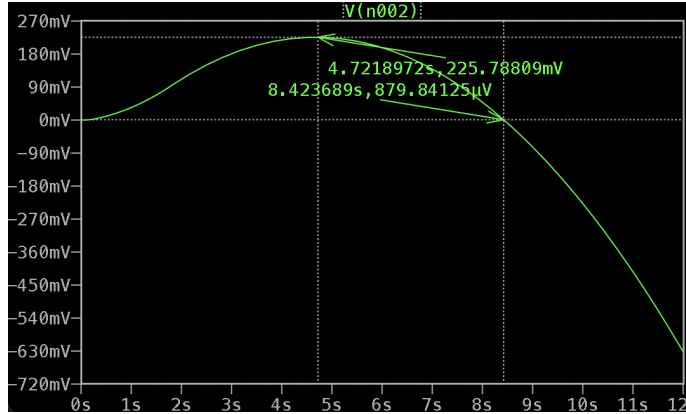


Figure 16: Part 4 Simulation

5 Part 5

All the experimental results are attached to the corresponding parts above.

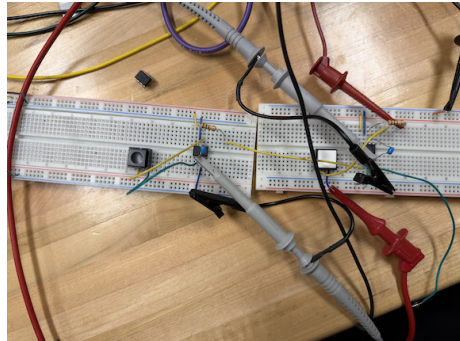


Figure 18: Our circuit. Left side is the second stage, right side is the first stage



Figure 19: Part 4 Projectile. Maximum height: ≈ 225.78 ft
Duration of flight: ≈ 10 s

6 Appendix

6.1 Thrust profiles

0.00 0.000 0.02 0.328 0.05 0.820 0.10 1.640 0.15 2.296 0.18 1.804 0.22 0.984 0.30
0.820 0.60 0.738 1.00 0.738 1.40 0.738 1.80 0.738 1.82 0.000 2.00 0.000